

Descriptive/Laboratory

1. A 0.01 M solution of a metal salt forms white precipitates when mixed with equal volumes of 0.01 M Na_2CO_3 and 0.01 M Na_2SO_4 , but no precipitate with 0.01 M NaOH . What is the metal salt?
- (A) MgCl_2 (B) ZnCl_2
(C) AgNO_3 (D) BaCl_2
2. A new balance is tested with replicate measurements of a standard 50.00 g mass, and gives readings of 49.31 g, 49.31 g, 49.29 g, and 49.30 g. Which description best characterizes the new balance?
- (A) Low accuracy and low precision
(B) Low accuracy and high precision
(C) High accuracy and low precision
(D) High accuracy and high precision
3. Oxalate ion, $\text{C}_2\text{O}_4^{2-}$, can be quantitated by titration with potassium permanganate in acidic solution. Which is used to identify the endpoint?
- (A) A persistent pink color from permanganate
(B) A persistent pink color from Mn^{2+}
(C) A persistent pink color from added phenolphthalein
(D) Bubbles of CO_2 are no longer observed
4. The number of moles of a weak monoprotic acid is determined by titrating with standardized NaOH solution to a phenolphthalein endpoint. The procedure calls for the acid to be dissolved in 50 mL of distilled water. What error would be introduced by using 75 mL of distilled water instead?
- (A) The number of moles would increase significantly.
(B) The number of moles would decrease significantly.
(C) The number of moles would not be significantly affected.
(D) The number of moles would be decreased by the change if the indicator concentration is low but increased if the indicator concentration is high.
5. Solutions of which metal salt give a yellow flame test?
- (A) Na (B) K (C) Cu (D) Ba
6. The concentration of which salt would be most difficult to measure using UV-visible spectroscopy?
- (A) NaMnO_4 (B) KClO_4
(C) $\text{Co}(\text{NO}_3)_2$ (D) CuCl_2
7. Which compound could be used to remove Ca^{2+} from hard water?
- (A) LiCl (B) Na_2CO_3
(C) KNO_3 (D) K_2SO_4
8. Addition of silver nitrate solution to 1.0 M solutions of which compounds will result in a yellow precipitate?
- I. NaCl II. KI
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
9. A student is cleaning a buret in preparation for a titration in which the buret will be filled with 0.100 M NaOH . What should be used for the final rinse of the buret?
- (A) Ethanol (B) Distilled water
(C) 0.100 M HCl (D) 0.100 M NaOH
10. In which solvent does solid iodine dissolve to give a purple solution?
- (A) Water (B) Ethanol
(C) Carbon tetrachloride (D) 0.1 M aqueous KI
11. Which glassware would be most suitable for preparing a 0.1500 M solution of silver nitrate?
- (A) 50 mL volumetric flask
(B) 50 mL round-bottom flask
(C) 50 mL Erlenmeyer flask
(D) 50 mL graduated cylinder
12. A 2.00 g sample of $\text{NaHCO}_3(s)$ is heated in a 10.00 g crucible to drive off H_2O and CO_2 . The mass data in the table were collected by heating the crucible and then periodically letting it cool to room temperature and reweighing it. Which is the first measurement after which heating could have been stopped?
- | Measurement | Mass of crucible and solid |
|-------------|----------------------------|
| 0 | 12.00 g |
| 1 | 11.48 g |
| 2 | 11.24 g |
| 3 | 11.25 g |
| 4 | 11.24 g |
- (A) Measurement 1 (B) Measurement 2
(C) Measurement 3 (D) Measurement 4

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13. Which apparatus would be most suitable for collecting a sample of solid barium sulfate produced through the reaction of aqueous solutions of barium hydroxide and sulfuric acid?
- (A) Barometer (B) Beaker
(C) Büchner funnel (D) Buret
14. A solution is prepared by dissolving 10.00 g of a salt in 90.00 g of water. What additional information is needed to calculate the molarity of this solution?
- I. The molar mass of the salt
II. The density of the solution
- (A) I alone would allow one to calculate the molarity.
(B) II alone would allow one to calculate the molarity.
(C) I and II together would allow one to calculate the molarity.
(D) The molarity cannot be calculated even if both I and II are known.
15. Which substance dissolves to give a colorless solution?
- (A) KClO_4 (B) K_2CrO_4
(C) KMnO_4 (D) $\text{K}_3\text{Fe}(\text{CN})_6$
16. Which substance does not give off heat when added to water?
- (A) NaOH (B) KNO_3
(C) H_2SO_4 (D) CaO
17. A solution of a nitrate salt does not form a precipitate when treated with dilute sulfuric acid. Which cation is not present in this solution?
- (A) NH_4^+ (B) Al^{3+}
(C) Fe^{2+} (D) Pb^{2+}
18. Which measurement would be least able to distinguish between a sample of elemental bromine prepared using exclusively the ^{79}Br isotope ($^{79}\text{Br}_2$) and one prepared using exclusively the ^{81}Br isotope ($^{81}\text{Br}_2$)?
- (A) Infrared spectroscopy (B) Liquid density
(C) Mass spectrometry (D) Vapor density
19. What is the main purpose of adding boiling chips to a solution that will be heated?
- (A) To reduce the boiling point of the solution
(B) To increase the boiling point of the solution
(C) To prevent the solution from superheating
(D) To decrease the time required for the solution to reach its boiling point
20. An aqueous solution contains one or more of the metal cations Mg^{2+} , Ba^{2+} , and Pb^{2+} at concentrations of 0.1 M. Addition of an equal volume of 0.1 M sodium sulfate causes formation of a precipitate. Which inference regarding the original solution is correct?
- (A) The solution must contain Mg^{2+} or Ba^{2+} .
(B) The solution must contain Mg^{2+} or Pb^{2+} .
(C) The solution must contain Ba^{2+} or Pb^{2+} .
(D) The solution must contain Mg^{2+} , Ba^{2+} , and Pb^{2+} .
21. Iron(III) nitrate is dissolved in water to form a 0.1 M Fe^{3+} solution and allowed to stand at room temperature for three days. During that time, a precipitate forms. Addition of which reagent will best prevent formation of a precipitate?
- (A) Sulfuric acid (B) Potassium hydroxide
(C) Sodium bicarbonate (D) Sodium chloride
22. Adding hydrogen peroxide to which reagent will NOT result in gas evolution?
- (A) Sodium hypochlorite (B) Copper(II) sulfate
(C) Manganese(IV) oxide (D) Sodium bisulfite
23. Acetylsalicylic acid is synthesized from salicylic acid, then isolated and dried. The melting point of the product is determined. If some salicylic acid is present, what will be observed? The melting point of pure acetylsalicylic acid is 136°C and the melting point of pure salicylic acid is 159°C .
- (A) The product will melt below 136°C .
(B) Most of the product will melt at 136°C , with some flecks of solid remaining until the temperature reaches 159°C , at which point the remaining material will melt.
(C) The product will melt between 136 and 159°C .
(D) The product will melt above 159°C .
24. A 10.00 g sample of an unknown metal is reacted with hydrochloric acid to form the divalent metal ion and hydrogen gas, which is then collected over water. The molar mass of the metal is calculated to determine the identity of the metal. Which will cause the calculated molar mass to be too large?
- I. Not correcting for the vapor pressure of water
II. Not allowing the reaction to go to completion
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

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25. What is the mobile phase in thin layer chromatography?

- (A) A liquid solvent such as isopropanol
- (B) A smooth solid surface such as a glass plate
- (C) A coating on a solid surface such as silica gel
- (D) A chamber such as a glass jar

26. Which is the strongest acid?

- (A) HBrO
- (B) HBrO_2
- (C) HBrO_3
- (D) HBrO_4

27. Which compound, when dissolved in water, will increase the electrical conductivity of water the least?

- (A) SO_3
- (B) H_2O_2
- (C) MgCl_2
- (D) NH_4Br

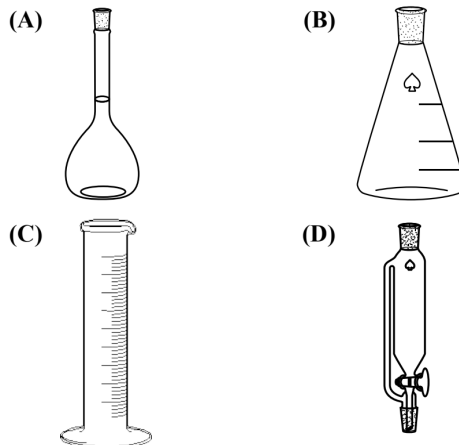
28. Which metal reacts most vigorously with water?

- (A) Aluminum (Al)
- (B) Zinc (Zn)
- (C) Cesium (Cs)
- (D) Mercury (Hg)

29. A student determines the acetic acid concentration of a sample of distilled vinegar by measuring 25.00 mL of the vinegar with a volumetric pipet, then titrating the sample with standardized sodium hydroxide solution using phenolphthalein as an indicator. Which error will give an acetic acid content for the vinegar that is too high?

- (A) The titration flask has some distilled water in it when the acetic acid sample is added to it.
- (B) The volumetric pipet has some distilled water in it when the acetic acid sample is measured with it.
- (C) Some of the vinegar remains in the volumetric pipet rather than being dispensed into the titration flask.
- (D) The endpoint is recorded when the solution turns dark red instead of faint pink.

30. Which 50-mL container would be most suitable for measuring and dispensing 37 mL of an aqueous solution?



31. Which salt gives a colorless aqueous solution?

- (A) NaClO_4
- (B) K_2CrO_4
- (C) $\text{Ni}(\text{NO}_3)_2$
- (D) CuSO_4

32. 100 mL of a 1.0 M solution of hydrochloric acid is gradually added to a stirred mixture containing 0.01 mol calcium hydroxide, 100 mL water, and a small amount of phenolphthalein. What changes are observed?

- I. The mixture changes from colorless to pink.
- II. The mixture changes from cloudy to clear.

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

33. Below is shown a section of a buret filled with titrant before beginning a titration. What initial volume should be recorded?



- (A) 1.20 mL
- (B) 1.26 mL
- (C) 2.74 mL
- (D) 2.80 mL

34. Titration with aqueous triiodide solution to a starch endpoint would be LEAST suitable for which determination?

- (A) Analysis of nickel content in a Ni/Co alloy
- (B) Determination of ascorbic acid in a vitamin C tablet
- (C) Measurement of sulfite concentration in a white wine
- (D) Determination of waters of hydration in a sodium thiosulfate sample

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35. Which liquid will NOT react with metallic sodium?
- (A) Water (B) Mineral oil
(C) Ethanol (D) Carbon tetrachloride
36. The molar enthalpy of solution of a salt that dissolves endothermically in water is measured in a coffee-cup calorimeter by weighing a known amount of distilled water into the cup and measuring its temperature, then adding a known mass of the salt to the water and measuring the temperature after the salt dissolves. If some water is initially present in the cup before the weighed amount of water is added, what is the effect on the experiment?
- (A) Calculated $\Delta H^\circ_{\text{soln}} > \text{True } \Delta H^\circ_{\text{soln}}$
(B) Calculated $\Delta H^\circ_{\text{soln}} = \text{True } \Delta H^\circ_{\text{soln}}$
(C) Calculated $\Delta H^\circ_{\text{soln}} < \text{True } \Delta H^\circ_{\text{soln}}$
(D) The reaction will be calculated to be exothermic even though it is actually endothermic.
37. When substances are separated by fractional distillation, which property is least typical of the substance that distills first?
- (A) Highest molar mass
(B) Weakest intermolecular forces
(C) Greatest vapor pressure
(D) Lowest boiling point
38. Which element is most abundant (by mass) in the Earth's crust?
- (A) Carbon (B) Oxygen
(C) Magnesium (D) Silicon
39. Which gas has the highest molar solubility in water at 25 °C and 1 atm?
- (A) CO₂ (B) NH₃ (C) O₂ (D) H₂S
40. Which compound is colorless?
- (A) NaMnO₄ (B) CrAsO₄
(C) RbIO₄ (D) BaCrO₄
41. Chlorine gas is bubbled into a colorless aqueous solution of sodium iodide. Which is the best description of what takes place?
- (A) A precipitate of white NaCl forms.
(B) A precipitate of metallic Na forms.
(C) The solution turns pale green as the chlorine dissolves.
(D) The solution turns yellow-brown as iodide reacts with the chlorine.
42. A student determines the number of moles of water in a hydrated metal oxide by weighing a clean, dry crucible and lid while the crucible is empty, then reweighing the crucible and lid with a sample of the hydrate, heating the crucible and lid with a Bunsen burner and then reweighing the crucible and lid with the sample after cooling to room temperature. Which error will result in too high a value for the amount of water of hydration?
- (A) The heating is conducted only once instead of the three times recommended by the procedure.
(B) The lid is left off the crucible when it is weighed with the hydrated oxide.
(C) The metal oxide reacts partially with oxygen in the air, forming a compound in a higher oxidation state.
(D) Some of the heated oxide is spilled from the crucible before it can be weighed.
43. Which salt dissolves in water to give a pink solution?
- (A) Co(CH₃COO)₂ (B) CuSO₄
(C) Fe(NO₃)₃ (D) ZnCl₂
44. A student is using a buret for a titration. What initial buret reading should be recorded?
- 
- (A) 6.6 mL (B) 7.4 mL
(C) 6.63 mL (D) 7.37 mL
45. A strip of metallic zinc is placed in a beaker containing dilute aqueous copper(II) nitrate. Which statement correctly describes what takes place?
- (A) No reaction takes place.
(B) The mass of the metal strip decreases as the zinc is oxidized.
(C) A white precipitate of CuNO₃ is formed.
(D) Bubbles of NO(g) form as the nitrate ion is reduced.

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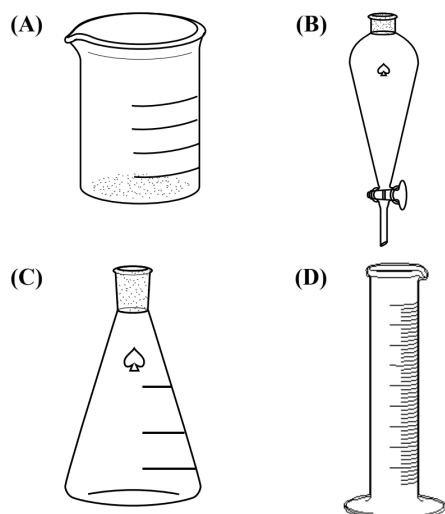
46. Which element does NOT have multiple allotropes?
- (A) Carbon (B) Oxygen
(C) Fluorine (D) Phosphorus
47. Which reaction is not exothermic?
- (A) Dilution of concentrated hydrochloric acid in water.
(B) Dilution of concentrated sulfuric acid in water.
(C) Dissolution of solid sodium hydroxide in water.
(D) Dissolution of solid sodium bicarbonate in water.
48. The molar mass of a volatile organic liquid is measured by weighing the mass of an empty flask of known volume, adding some of the liquid, heating the flask in a water bath until the liquid has just vaporized completely, then sealing the flask, letting it cool, and remeasuring the mass of the flask with the remaining organic compound. Which error will lead to a calculated value of the molar mass that is lower than the theoretical value?
- (A) The flask is not sealed promptly after the last of the compound vaporizes.
(B) The organic compound dimerizes to an appreciable extent in the gas phase.
(C) The barometric pressure was assumed to be 1 atm but in fact was greater than 1 atm.
(D) The volume of the flask used in the calculation is smaller than the true volume.
49. A solution of which compound gives a violet flame test?
- (A) LiBr (B) NaCl
(C) KNO₃ (D) B(OH)₃
50. Addition of 1 M HBr to 0.1 M solutions of which compound results in evolution of a colorless gas?
- (A) NaHSO₃ (B) NaHSO₄
(C) Ce(NH₄)₂(NO₃)₆ (D) KBrO₃
51. Which solid is least soluble in water at 298 K?
- (A) CaF₂ (B) AgF
(C) Ba(OH)₂ (D) CoSO₄
52. Which methods would be suitable for determining the concentration of an aqueous solution of KMnO₄?
- I. Visible spectrophotometry (colorimetry)
II. Redox titration
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
53. The concentration of sulfate ion in a solution is measured by precipitating the sulfate as BaSO₄, filtering the precipitate on ashless filter paper, and heating the filter paper and precipitate in a tared crucible with a Bunsen burner. Which error will result in a sulfate concentration that is higher than the actual concentration?
- (A) The empty crucible contains a few drops of water when it is tared.
(B) A glass fiber filter is used instead of ashless filter paper.
(C) Some fine precipitate is not captured by the filter.
(D) Some of the sulfate-containing solution spills before the BaCl₂ solution is added.
54. Which is the best way to dispense liquids using a volumetric pipet?
- (A) The pipet is immersed in the liquid to be dispensed, then lifted with a gloved finger on the top and the liquid allowed to drain to the mark. The remaining contents are then allowed to drain into the desired container.
(B) The pipet is immersed in the liquid to be dispensed, then lifted with a gloved finger on the top and the liquid allowed to drain to the mark. The remaining contents are then allowed to drain into the desired container, with a pipet bulb used to gently blow out any residual droplets.
(C) The tip of the pipet is submerged below the surface of the liquid and suction is applied using a pipet bulb until the liquid rises above the level of the mark. A gloved finger is then applied to the top of the pipet and the liquid allowed to drain to the mark. The remaining contents are then allowed to drain into the desired container.
(D) The tip of the pipet is submerged below the surface of the liquid and suction is applied by mouth until the liquid rises to the level of the mark. The contents are then allowed to drain into the desired container.
55. Addition of 6 M HCl to which substance will NOT result in gas evolution?
- (A) Al (B) Zn (C) K₂CO₃ (D) NaNO₃
56. A divalent metal ion dissolved in dilute hydrochloric acid forms a precipitate when H₂S is bubbled through the solution. Which ion is it?
- (A) Ca²⁺ (B) Mn²⁺ (C) Zn²⁺ (D) Cd²⁺
57. Which compound will form the most intensely colored 0.01 M aqueous solution?
- (A) KMnO₄ (B) KClO₄
(C) KAl(SO₄)₂ (D) KI

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58. A solution contains 0.1 M Sr^{2+} ions and 0.1 M Ag^+ ions. Addition of an equal volume of a 0.5 M solution of which reagent will cause precipitation of a strontium salt but not a silver salt?
- (A) NaNO_3 (B) NaF
(C) NaOH (D) NaCl
59. Which would be most suitable for measuring 2.7 mL of ethanol for addition to a reaction with acidified dichromate?
- (A) 10-mL graduated cylinder
(B) 10-mL volumetric flask
(C) 10-mL volumetric pipet
(D) 10-mL beaker
60. The molar mass of a solid carboxylic acid is determined by titrating a known mass of the acid with a standardized solution of NaOH to a phenolphthalein endpoint. Which errors will lead to a molar mass that is smaller than the actual molar mass?
- I. Some of the acid is spilled when being transferred into the titration flask.
II. The endpoint is recorded when the solution is dark red in color rather than light pink.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
61. A white ionic solid is dissolved in water. Addition of a solution of sodium chloride to this solution results in a white precipitate. What was the cation in the original ionic solid?
- (A) Na^+ (B) Fe^{3+} (C) Ag^+ (D) Sr^{2+}
62. An element is a solid at room temperature but soft enough to be cut with an ordinary knife. When placed in water, the element reacts violently. What element is it?
- (A) Na (B) Mg (C) Cu (D) Hg
63. A 0.1 M solution of which salt is the most basic?
- (A) NaNO_3 (B) NaClO_4
(C) NaHSO_4 (D) NaHCO_3
64. Addition of small amounts of which solids to 4 M HCl will result in gas evolution?
- I. Zn II. Na_2SO_3
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
65. A student determines the acetic acid concentration of a sample of distilled vinegar by titration of 25.00 mL of the vinegar with standardized sodium hydroxide solution using phenolphthalein as an indicator. Which error will give an acetic acid content for the vinegar that is too low?
- (A) Some of the vinegar is spilled when being transferred from the volumetric flask to the titration flask.
(B) The NaOH solution is allowed to stand for a prolonged period after standardization and absorbs carbon dioxide from the air.
(C) The endpoint is recorded when the solution turns dark red instead of faint pink.
(D) The vinegar is diluted with distilled water in the titration flask before the NaOH solution is added.
66. Which piece of equipment would give the most precise delivery of 25.0 mL of a solution?
- (A) 25-mL graduated cylinder
(B) 25-mL syringe
(C) 25-mL beaker
(D) 25-mL volumetric pipet
67. Dissolution of which salt in water results in a decrease in the temperature of the solution?
- (A) KHSO_4 (B) NaOH
(C) AlCl_3 (D) NH_4NO_3
68. What is observed when equal volumes of 0.1 M aqueous HCl and 0.01 M aqueous Na_2SO_3 are mixed?
- (A) Colorless solution and a white precipitate
(B) Colored solution and a white precipitate
(C) Colorless solution and a colored precipitate
(D) Colorless solution, no precipitate, and gas evolution
69. Which combination of dilute aqueous reagents will not produce a precipitate?
- (A) $\text{AgNO}_3 + \text{HCl}$ (B) $\text{NaOH} + \text{HClO}_4$
(C) $\text{BaBr}_2 + \text{Na}_2\text{SO}_4$ (D) $\text{ZnI}_2 + \text{KOH}$
70. A solution of a salt of which metal produces a bright red color in a flame test?
- (A) Lithium (B) Sodium
(C) Potassium (D) Copper
71. Which cation forms a colorless aqueous solution?
- (A) Co^{2+} (B) Ni^{2+} (C) Cu^{2+} (D) Zn^{2+}

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72. Which 50-mL container would be most suitable for measuring and dispensing 37 mL of an aqueous solution?



73. Which calcium compound is not appreciably more soluble in 0.1 M hydrochloric acid than it is in pure water?

- (A) Limestone, CaCO_3
(B) Slaked lime, Ca(OH)_2
(C) Gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
(D) Hydroxyapatite, $\text{Ca}_5(\text{OH})(\text{PO}_4)_3$

74. Which metal reacts most vigorously with water at 25 °C?

- (A) Na (B) Mg (C) K (D) Ca

75. Each of the following forms a colored aqueous solution EXCEPT

- (A) $\text{Cr(NO}_3)_3$ (B) $\text{Co(NO}_3)_2$
(C) $\text{Cu(NO}_3)_2$ (D) $\text{Zn(NO}_3)_2$

76. A student wishes to measure 37 mL of a liquid. Which apparatus would be most suitable?

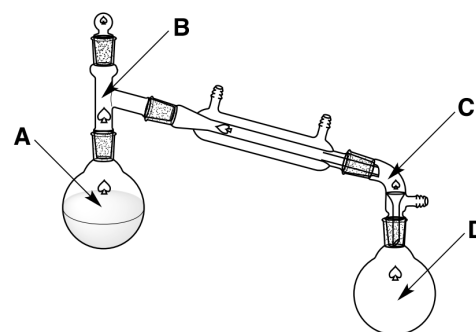
- (A) 50 mL graduated cylinder
(B) 50 mL volumetric pipet
(C) 50 mL beaker
(D) 50 mL Erlenmeyer flask

77. A 2.0 mL sample of a colorless solution, when treated with a few drops of 2 M hydrochloric acid, forms a white precipitate which dissolves when the solution is heated to boiling. The original solution could have contained which of the following cations?

- I. 0.1 M Ag^+ II. 0.1 M Pb^{2+}

- (A) I only (B) II only
(C) Either I or II (D) Neither I nor II

78. A student is separating CHCl_3 (bp = 61 °C) from $\text{CHCl}_2\text{CHCl}_2$ (bp = 146 °C) by distillation. She has just begun to collect the first distillate in the receiving flask. At what position in the apparatus will the temperature be 61 °C?



- (A) A (B) B (C) C (D) D

79. Which list gives nonmetals that are found in their elemental forms in nature?

- (A) neon, phosphorus, fluorine
(B) helium, hydrogen, iodine
(C) nitrogen, oxygen, sulfur
(D) oxygen, chlorine, phosphorus

80. Which one of the following cannot act as an oxidizing agent?

- (A) S^{2-} (B) SO_3^{2-} (C) SO_4^{2-} (D) $\text{S}_2\text{O}_8^{2-}$

81. Which substance is the primary component in stalactites and stalagmites in caves?

- (A) CaO (B) CaCO_3 (C) Ca(OH)_2 (D) CaSO_4

82. Five successive determinations of the density of an alloy gave the following results:
10.29 g/mL, 9.95 g/mL, 10.06 g/mL, 9.89 g/mL, 10.18 g/mL
What value should be reported for the density of this alloy?

- (A) 10.074 g/mL (B) 10.07 ± 0.16 g/mL
(C) 10.1 ± 0.2 g/mL (D) 1.0×10^1 g/mL

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83. A student is asked to analyze a water sample from a stream for total solids (TS), dissolved solids (DS), and suspended solids (SS). She carries out the experiments below.
1. A 25-mL portion of the water sample is evaporated to dryness in a pre-weighed evaporating dish to give mass 1.
 2. A separate 25-mL portion is filtered into a second pre-weighed evaporating dish and evaporated to dryness to give mass 2.

How are values for TS, SS and DS (per 25 mL water) determined?

- (A) TS = mass 1, SS = mass 1 – mass 2, DS = mass 2
 (B) TS = mass 1, SS = mass 2, DS = mass 1 – mass 2
 (C) TS = mass 1 + mass 2, SS = mass 1, DS = mass 2
 (D) TS = mass 1 + mass 2, SS = mass 2, DS = mass 1

84. Which method(s) can be used to determine the concentration of HNO_3 in an aqueous solution of nitric acid?

I titration with a standard base
 II titration with a standard oxidizing agent
 III precipitation with Ag^+

- (A) I only (B) III only
 (C) I and II only (D) I, II, and III

85. A solid can be separated from a liquid by all the following means EXCEPT

- (A) decantation (B) distillation
 (C) filtration (D) hydration

86. A student determined the density of a solid to be 2.90, 2.91 and 2.93 $\text{g}\cdot\text{cm}^{-3}$. If the actual density of this solid is 2.70 $\text{g}\cdot\text{cm}^{-3}$, how should the student's results be described?

- (A) high accuracy and high precision
 (B) low accuracy and high precision
 (C) high accuracy and low precision
 (D) low accuracy and low precision

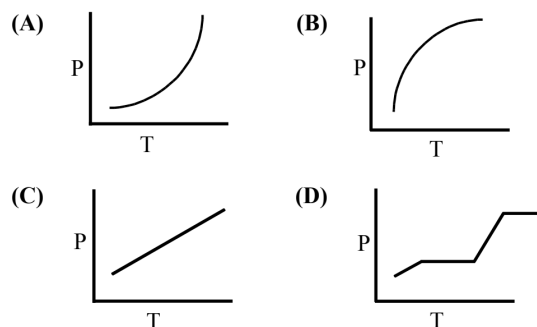
87. Which cation forms an insoluble chloride and an insoluble sulfide?

- (A) Ba^{2+} (B) Cu^{2+} (C) Mn^{2+} (D) Pb^{2+}

88. Which 0.10 M aqueous solution exhibits the lowest electrical conductivity?

- (A) $\text{HC}_2\text{H}_3\text{O}_2(\text{aq})$ (B) $\text{HNO}_3(\text{aq})$
 (C) $\text{NH}_4\text{C}_2\text{H}_3\text{O}_2(\text{aq})$ (D) $\text{Ca}(\text{NO}_3)_2(\text{aq})$

89. Which graph best represents the vapor pressure of water as a function of temperature from 0 °C to 100 °C?



90. The table below shows the data for three titrations to determine the concentration of a NaOH solution with standard 0.200 M HCl solution using phenolphthalein as the indicator.

Trial	Vol HCl, mL	Vol NaOH, mL	$M_{\text{NaOH, calc.}}$
1	21.43	19.26	0.223
2	18.57	16.73	0.222
3	22.20	21.14	0.210

Which explanation best accounts for the lower value of the NaOH M in Trial 3?

- (A) Some of the neutralized solution from Trial 2 was left in the flask for Trial 3.
 (B) The number of drops of phenolphthalein was doubled in Trial 3.
 (C) The HCl concentration was used as 0.250 M in the NaOH molarity calculation.
 (D) A few drops of NaOH solution were spilled on the desktop in Trial 3.
91. What is the characteristic color of the flame test for potassium?
 (A) yellow (B) red (C) green (D) violet
92. A dilute solution of which acid is most likely to produce a reduction product other than H_2 when it reacts with a metal?
 (A) HF (B) HCl (C) HNO_3 (D) H_2SO_4
93. Which two substances react spontaneously?
 (A) Ag and Cu (B) Ag^+ and Cu
 (C) Ag and Cu^{2+} (D) Ag^+ and Cu^{2+}

Descriptive/Laboratory

94. Which piece of laboratory apparatus is not used for the purpose listed?

	Apparatus	Purpose
(A)	aspirator	measure height
(B)	buret	measure volume
(C)	calorimeter	measure thermal energy change
(D)	desiccator	store samples in dry conditions

(A) A (B) B (C) C (D) D

95. Which transformation demonstrates that the bonds between water molecules are weaker than the bonds within a water molecule?

(A) freezing water (B) electrolysis of water
(C) boiling water (D) reaction of water with Na(s)

96. The molar mass of an unknown organic liquid ($M \sim 100$) is determined by placing 5 mL of the liquid in a weighed 125-mL conical flask fitted with a piece of Al foil with a pin hole in it. The flask is heated in a boiling water bath until the liquid evaporates to expel the air and fill the flask with the unknown vapor at atmospheric pressure. After cooling to room temperature the flask and its contents are reweighed. The uncertainty in which piece of apparatus causes the largest percentage error in the molar mass?

(A) balance (± 0.01 g) (B) barometer (± 0.2 mm Hg)
(C) flask (± 1.0 mL) (D) thermometer (± 0.2 °C)

97. A student is asked to dispense 24.70 mL of a solution with an uncertainty of less than 0.05 mL. Which item should be used for this task?

(A) 50 mL beaker
(B) 50 mL buret
(C) 50 mL Erlenmeyer flask
(D) 50 mL graduated cylinder

98. Oxygen gas can be produced by the decomposition of all of the following substances EXCEPT

(A) calcium oxide. (B) hydrogen peroxide.
(C) mercury(II) oxide. (D) ozone.

99. Which gaseous product is formed when dilute nitric acid reacts with silver metal in the absence of air?

(A) H_2 (B) O_2 (C) NH_3 (D) NO

100. How many of the following gases are characterized by BOTH color AND a distinctive odor?

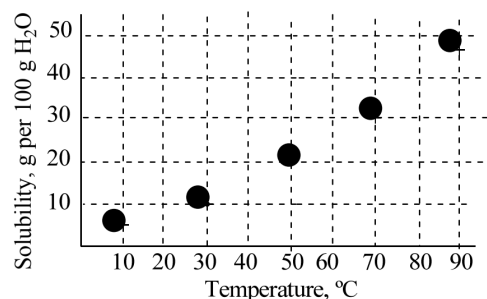
Cl_2 CH_4 NO_2

(A) none (B) one (C) two (D) three

101. Which action should be taken immediately if concentrated sulfuric acid is spilled on the skin?

(A) It should be rinsed off with large quantities of running water.
(B) It should be neutralized with solid $CaCO_3$.
(C) It should be neutralized with concentrated NaOH.
(D) The area of the spill should be wrapped tightly with cloth and shown to a health provider.

102. The solubility of $KClO_3$ at several temperatures is shown in the accompanying diagram.



A student mixes 10.0 g of $KClO_3$ with 45.0 g of H_2O and stirs it for a long time at 60 °C until the solution is completely clear then allows it to cool slowly to 20 °C where it remains clear. Which statement about the final clear mixture at 20 °C is correct?

(A) It is a saturated solution.
(B) It is an unsaturated solution and can be made saturated by decreasing the temperature.
(C) It is an unsaturated solution and can be made saturated by increasing the temperature.
(D) It is a supersaturated solution.

103. A student is asked to measure 30.0 g of methanol ($d = 0.7914$ g/mL at 25 °C) but has only a graduated cylinder with which to measure it. What volume of methanol should the student use to obtain the required 30.0 g?

(A) 23.7 mL (B) 30.0 mL
(C) 32.4 mL (D) 37.9 mL

104. A flame test was performed to confirm the identity of a metal ion in solution. The result was a green flame. Which of the following metal ions is indicated?

(A) copper (B) sodium
(C) strontium (D) zinc

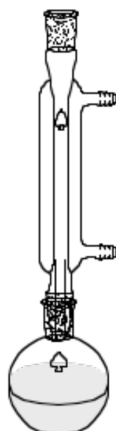
Descriptive/Laboratory

105. When phenolphthalein is added to an aqueous solution containing one of the following solutes the solution turns pink. Which solute is present?
- (A) NaCl (B) $\text{KC}_2\text{H}_3\text{O}_2$
(C) LiBr (D) NH_4NO_3
106. Solid camphor is insoluble in water but is soluble in vegetable oil. The best explanation for this behavior is that camphor is a(n)
- (A) ionic solid (B) metallic solid
(C) molecular solid (D) network solid
107. A student performed an experiment to determine the ratio of H_2O to CuSO_4 in a sample of hydrated copper(II) sulfate by heating it to drive off the water and weighing the solid before and after heating. The formula obtained experimentally was $\text{CuSO}_4 \cdot 5.5\text{H}_2\text{O}$ but the accepted formula is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. Which error best accounts for the difference in results?
- (A) During heating some of the hydrated copper(II) sulfate was lost.
(B) The hydrated sample was not heated long enough to drive off all the water.
(C) The student weighed out too much sample initially.
(D) The student used a balance that gave weights that were consistently too high by 0.10 g.
108. An aqueous solution is known to contain Ag^+ , Mg^{2+} , and Sr^{2+} ions. Which reagent should be used to selectively precipitate the Ag^+ ?
- (A) 0.20 M NaCl (B) 0.20 M NaOH
(C) 0.20 M Na_2SO_4 (D) 0.20 M Na_3PO_4
109. Which salt is colorless?
- (A) KMnO_4 (B) BaSO_4
(C) Na_2CrO_4 (D) CoCl_2
110. Which 0.10 M aqueous solution exhibits the lowest electrical conductivity?
- (A) NH_4Cl (B) CuBr_2
(C) Na_2CO_3 (D) $\text{C}_2\text{H}_5\text{OH}$
111. Which element is a liquid at 25°C and 1 atm?
- (A) fluorine (B) chlorine
(C) bromine (D) iodine
112. Mixing which combination produces a gaseous product?
- (A) solid ammonium nitrate and solid calcium hydroxide
(B) copper metal and 0.10 M hydrochloric acid
(C) solutions of barium hydroxide and 0.10 M sulfuric acid
(D) solutions of aluminum nitrate and sodium chloride
113. Which technique can be used to determine the number of components in a plant pigment?
- (A) calorimetry (B) chromatography
(C) colorimetry (D) gravimetry
114. In the determination of the molar mass of a solid acid by titrating it with a standardized base, which procedural error will yield a molar mass that is smaller than the actual value?
- (A) adding the standardized base to a buret containing drops of water
(B) dissolving the weighed solid acid in twice the recommended volume of water
(C) using half as many drops of indicator as suggested
(D) weighing out half of the recommended mass of solid acid
115. Which element is a liquid at 25°C and 1.0 atm?
- (A) bromine (B) krypton
(C) phosphorus (D) xenon
116. Which pair of aqueous solutions produce a yellow precipitate upon mixing?
- (A) AlCl_3 and KOH (B) $\text{Ba}(\text{NO}_3)_2$ and Na_2SO_4
(C) $\text{Cu}(\text{NO}_3)_2$ and NaClO_4 (D) $\text{Pb}(\text{C}_2\text{H}_3\text{O}_2)_2$ and KI
117. A student wishes to determine the thickness of a rectangular piece of aluminum foil but cannot measure it directly. She can measure its density (d), length (l), mass (m) and width (w). Which relationship will give the thickness?
- (A) $\frac{m}{d \cdot l \cdot w}$ (B) $\frac{m \cdot l \cdot w}{d}$
(C) $\frac{d \cdot l \cdot w}{m}$ (D) $\frac{d \cdot m}{l \cdot w}$

Descriptive/Laboratory

118. The apparatus shown would be used to

- (A) distill a liquid.
- (B) reflux a solution.
- (C) filter a precipitate.
- (D) chromatograph a mixture.



119. Which element is the major component in solar cells?

- (A) As (B) Ge (C) P (D) Si

120. Which procedure is recommended for reading the level of an aqueous solution in a buret or graduated cylinder?

- (A) Keep the eye level with the bottom of the meniscus and record that reading.
- (B) Keep the eye level with the top of the meniscus and record that reading.
- (C) Look down at the meniscus at an angle to obtain the average reading directly.
- (D) Read both the bottom and top of the meniscus and average those readings.

121. Which compound is most soluble in water?

- (A) AgCl (B) Ag₂CO₃ (C) BaCl₂ (D) BaCO₃

122. Which piece of laboratory equipment should be used to deliver a 10.00 mL sample of acid from a stock container to a flask for a titration?

- (A) 1.0 mL Beral pipet used 10 times
- (B) 10 mL graduated cylinder
- (C) 10 mL volumetric pipet
- (D) 25 mL beaker

123. When solid KOH is mixed with solid NH₄Cl a gas is produced. Which gas is it?

- (A) Cl₂ (B) H₂ (C) HCl (D) NH₃

124. An experiment is carried out to determine the molar mass of a compound by the freezing point depression method using the equation

$$MM = 7.05 \frac{\text{mass solute}}{\Delta T \times \text{kg solvent}}$$

The data below are collected.

Mass of empty test tube	42.0 g
Mass of test tube and solvent	73.6 g
Mass of solute dissolved in solvent	2.000 g
Freezing point of pure solvent	78.1 °C
Freezing point of solution	77.6 °C

How many significant figures can be reported for the molar mass of the solute?

- (A) 1 (B) 2 (C) 3 (D) 4

125. Which compound forms a colorless solution when dissolved in H₂O?

- (A) Co(NO₃)₂ (B) KMnO₄
(C) Na₂Cr₂O₇ (D) ZnCl₂

126. If an individual spills some 8 M H₂SO₄ on her/his arm, what treatment should be used?

- (A) Neutralize it immediately with a paste of NaOH in H₂O.
- (B) Rinse it with H₂O followed by a dilute solution of NaHCO₃.
- (C) Wash it with a solution of concentrated aqueous NH₃.
- (D) Wrap it tightly with gauze coated with petroleum jelly.

127. Which gas should not be collected over water because of its high solubility in water?

- (A) H₂ (B) CO (C) CH₄ (D) HCl

128. A student wishes to measure 17.3 mL of a standard solution. This can be done best using which container?

- (A) 25 mL beaker (B) 25 mL volumetric pipet
(C) 25 mL buret (D) 25 mL volumetric flask

129. In a familiar classroom demonstration, concentrated H₂SO₄ is added to a beaker containing sucrose (C₁₂H₂₂O₁₁), to produce a column of carbon. In this reaction the H₂SO₄ is acting primarily as a

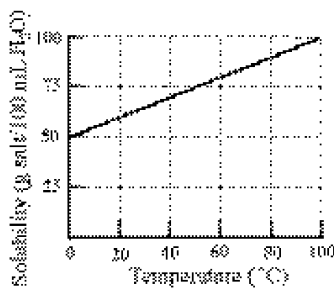
- (A) complexing agent. (B) dehydrating agent.
(C) oxidizing agent. (D) precipitating agent.

Descriptive/Laboratory

130. Aqua regia, the reagent that can be used to dissolve gold, is a 3:1 mixture of which acids?

(A) hydrochloric and sulfuric acids
(B) hydrofluoric and nitric acids
(C) hydrochloric and nitric acids
(D) perchloric and sulfuric acids

131. The solubility of a solid in H_2O at different temperatures is indicated in the accompanying diagram. What mass of the solid will crystallize when 40. mL of a solution that is saturated at 80°C is cooled to 20°C ?



(A) 12 g (B) 24 g (C) 30 g (D) 36 g

132. A student is asked to determine the degree of hydration of a salt by finding the mass of a clean dried crucible and lid followed by the mass of the crucible and lid containing a sample of the salt. The crucible, lid and salt are heated in a Bunsen burner flame, cooled, and weighed. If the crucible, lid and salt are heated only once instead of the recommended three times, what will be the effect on the degree of hydration compared with the actual value?

(A) too high (B) too low
(C) unaffected (D) impossible to predict

133. When 6 M hydrochloric acid is added to an unknown white solid, a colorless gas is produced. What is a possible identity for this solid?

(A) calcium nitrate (B) copper(II) chloride
(C) potassium sulfate (D) sodium carbonate

134. What is the first change that occurs when $\text{I}_2(\text{s})$ is heated slowly at one atmosphere pressure?

(A) The solid melts.
(B) The solid vaporizes.
(C) The solid breaks into atoms.
(D) The solid becomes darker in color.

135. Two pure organic compounds melt at 112°C and 114°C , respectively. If equal quantities of them are mixed, at what temperature will the mixture begin to melt?

(A) below 112°C (B) at 112°C
(C) between 112°C and 114°C (D) above 114°C

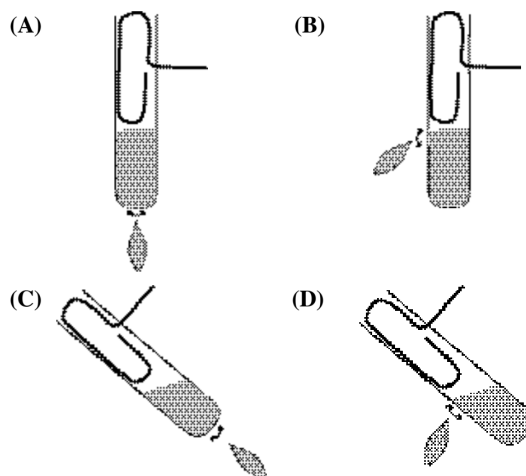
136. What is the major reason for using mercury (rather than water) in barometers?

(A) Mercury is much denser than water.
(B) Mercury has a higher boiling point than water.
(C) Mercury is chemically unreactive compared with water.
(D) Mercury expands with a decrease in air pressure; water does not.

137. How can 0.1 g samples of the two white solids, lead(II) chloride and silver chloride, be distinguished from one another?

(A) Add 10 mL of cold water to each. The silver chloride will dissolve.
(B) Add 10 mL of hot water to each. The lead(II) chloride will dissolve.
(C) Add 10 mL of sodium chloride to each solution. The lead(II) chloride will become warm and release chlorine gas.
(D) Add 10 mL of zinc chloride solution to each. The silver chloride will change to metallic silver.

138. Which is the proper way to heat a liquid in a test tube?



139. Which element is a gas at 25°C and 1 atm pressure?

(A) chlorine (B) phosphorus
(C) silicon (D) sulfur

140. Which combustion product is produced THE LEAST by gasoline-powered vehicles?

(A) CO_2 (B) H_2O (C) NO_2 (D) SO_2

141. Which element has the highest electrical conductivity at room temperature?

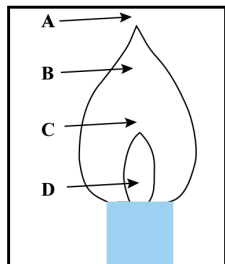
(A) Ge (B) Se (C) Sn (D) Te

Descriptive/Laboratory

142. How should a student prepare 100 mL of a 1.0 M H_2SO_4 solution from a 10. M H_2SO_4 solution?

- (A) Add 90 mL of H_2O to 10 mL of 10 M H_2SO_4 .
- (B) Add 10 mL of 10 M H_2SO_4 to 90 mL of H_2O .
- (C) Add 10 mL of 10 M H_2SO_4 to 80 mL of H_2O , stir and dilute to 100 mL after allowing to cool.
- (D) Add 80 mL of H_2O to 10 mL of 10 M H_2SO_4 , stir and dilute to 100 mL after allowing to cool.

143. Which letter in the diagram depicts the hottest portion of a Bunsen burner flame?



- (A) A (B) B (C) C (D) D

144. What is the proper technique to test the odor of a vapor in a test tube?

- (A) Hold the test tube near the nose and sniff.
- (B) Use a micropipet to capture some of the gas and sniff that.
- (C) Hold the test tube above the nose and pour the gas toward it.
- (D) Hold the test tube near the nose and waft the gas toward it with a hand.

145. Which anion forms the smallest number of insoluble salts?

- (A) Cl^- (B) NO_3^- (C) CO_3^{2-} (D) SO_4^{2-}

146. Which piece of apparatus can measure a volume of 25.0 mL most precisely?

- (A) 25 mL beaker
- (B) 25 mL conical flask
- (C) 25 mL graduated cylinder
- (D) 25 mL pipet

147. How many significant figures should be reported in the answer to the calculation (Assume all numbers are experimentally determined.)

$$\frac{12.501 \times 3.52}{0.0042} + 6.044$$

- (A) 2 (B) 3 (C) 4 (D) 5

148. Five pellets of a metal have a total mass of 1.25 g and a total volume of 0.278 mL. What is the density of the metal in $\text{g}\cdot\text{mL}^{-1}$?

- (A) 0.348 (B) 0.900 (C) 4.50 (D) 22.5

149. What is the color of the flame test for sodium?

- (A) green (B) red
- (C) violet (D) yellow

150. When is it acceptable to eat in a chemistry laboratory?

- (A) Anytime when a person is not doing an experiment.
- (B) Whenever there are no hazardous chemicals out.
- (C) If it is necessary to do so in order to keep another appointment.
- (D) Never.

151. Which element reacts most rapidly with water at 25 °C to produce a gas?

- (A) aluminum (B) carbon
- (C) lithium (D) phosphorus

152. Which is the best procedure to follow if a student spills several drops of concentrated HCl on his hand?

- (A) Cover the area with solid sodium hydrogen carbonate.
- (B) Rinse with large amounts of cold water.
- (C) Wash with concentrated sodium hydroxide solution.
- (D) Wrap the hand with sterile gauze.

153. Which pair of substances can be combined to produce ammonia gas?

1. $(\text{NH}_4)_2\text{SO}_4(s)$ and $\text{NaOH}(aq)$

2. $\text{NH}_3(aq)$ and $\text{HCl}(aq)$

- (A) 1 only (B) 2 only
- (C) both 1 and 2 (D) neither 1 nor 2

154. What products result when equal volumes of equimolar aqueous solutions of copper(II) sulfate and barium hydroxide are mixed?

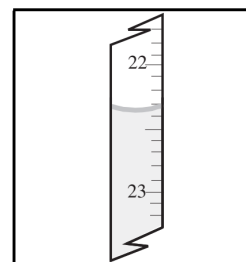
- (A) $\text{Ba}^{2+}(aq)$, $\text{Cu}^{2+}(aq)$, $\text{OH}^-(aq)$, and $\text{SO}_4^{2-}(aq)$
- (B) $\text{Cu}(\text{OH})_2(s)$, $\text{Ba}^{2+}(aq)$, and $\text{SO}_4^{2-}(aq)$
- (C) $\text{BaSO}_4(s)$, $\text{Cu}^{2+}(aq)$, and $\text{OH}^-(aq)$
- (D) $\text{BaSO}_4(s)$ and $\text{Cu}(\text{OH})_2(s)$

155. Which statement about silicon is *false*?

- (A) It is a metalloid.
- (B) It behaves as a semiconductor when pure.
- (C) It is extremely rare in the earth's crust.
- (D) It has a smaller atomic radius than aluminum.

Descriptive/Laboratory

156. A standard HCl solution is titrated to a pink phenolphthalein endpoint by adding a NaOH solution while stirring. If a solution becomes pink throughout but loses its color upon standing for a short time, what should be done to restore the color?
- (A) Add more phenolphthalein indicator.
(B) Add an additional drop of NaOH solution.
(C) Add an additional drop of HCl solution.
(D) Stir more vigorously.
157. A sample of gas in a small test tube produces a pop when a burning splint is inserted. Which gas could it be?
- (A) H_2 (B) O_2 (C) Cl_2 (D) NO
158. Electrolysis is used commercially to isolate which metal(s)?
1. Al 2. Fe
- (A) 1 only (B) 2 only
(C) both 1 and 2 (D) neither 1 nor 2
159. Which metal reacts most vigorously with water?
- (A) Al (B) Ca (C) Fe (D) K
160. Which are strong acids?
- I. HI II. HNO_3 III. H_2SO_3
- (A) II only (B) I and II only
(C) II and III only (D) I, II and III
161. What products result when aqueous solutions of CuCl_2 and $(\text{NH}_4)_2\text{S}$ are mixed?
- (A) CuS(aq) and $\text{NH}_4\text{Cl(s)}$ (B) CuS(s) and $\text{NH}_4\text{Cl(aq)}$
(C) CuS(aq) and $\text{NH}_4\text{Cl(g)}$ (D) CuS(s) and $\text{NH}_4\text{Cl(s)}$
162. Which element has the highest electrical conductivity?
- (A) Ga (B) Ge (C) As (D) Si
163. The molar mass of a volatile liquid (bp $< 90^\circ\text{C}$) is to be determined by measuring the density of its vapor in an Erlenmeyer flask capped with Al foil with a pinhole to allow the vapor to escape. A sample of the liquid is added to the pre-weighed flask that is heated in a water bath until the liquid has evaporated, after which the flask is dried and reweighed. Which piece of information is *not* required to determine the molar mass of the liquid?
- (A) barometric pressure (B) mass of liquid sample
(C) temperature of water (D) volume of the flask
164. Which set of reactants produces a gaseous product?
- I. $6 \text{ M HNO}_3(\text{aq}) + \text{Cu(s)}$ II. $6 \text{ M HCl(aq)} + \text{CaCO}_3(\text{s})$
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
165. Which metal, when exposed to air, forms an oxide coating that protects the metal from further reaction?
- (A) Al (B) Cu (C) Fe (D) Zn
166. The principal reason that solid sodium hydroxide is *not* used as a primary standard for acid-base titrations is that it
- (A) absorbs water from air.
(B) has a low molar mass.
(C) reacts slowly with many acids.
(D) ionizes in water.
167. Which metal reacts most vigorously with water?
- (A) Ca (B) K (C) Mg (D) Na
168. Which substance produces an acidic solution when it is bubbled into water?
- (A) CO_2 (B) Ar (C) NH_3 (D) CH_4
169. Which substance is the least soluble in H_2O ?
- (A) K_2CO_3 (B) KHCO_3
(C) $\text{Ca}(\text{HCO}_3)_2$ (D) CaCO_3
170. What value should be reported for the buret reading shown?



- (A) 22.3 mL (B) 22.30 mL
(C) 22.36 mL (D) 22.40 mL

Descriptive/Laboratory

171. Which technique is recommended for determining the odor of an unknown liquid in the laboratory?
- (A) Hold a test tube of the liquid under the nose and inhale the vapor.
 - (B) Transfer a few drops of the liquid to the desk top and inhale the vapor from there.
 - (C) Use a medicine dropper to collect a small quantity of the vapor and squirt this under the nose while inhaling.
 - (D) Use a hand to fan some of the vapor from the test tube to the nose and inhale.
172. A colorless solution is known to contain one of these ions. Which ion is present if adding dilute HCl produces a white precipitate that dissolves when the solution is warmed?
- (A) Ag^+ (B) Cu^{2+} (C) Hg_2^{2+} (D) Pb^{2+}
173. A student is asked to measure 12 mL of a liquid as precisely as possible. Which piece of equipment should she select for this task?
- (A) 25 mL beaker
 - (B) 25 mL graduated cylinder
 - (C) 25 mL conical flask
 - (D) 25 mL volumetric flask
174. Which separation technique is based on differences in the volatility of the substances to be separated?
- (A) filtration (B) distillation
- (C) solvent extraction (D) paper chromatography
175. Which gas is least suitable for collection over water?
- (A) Ar (B) O_2 (C) CO_2 (D) NH_3